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Strawberry Disease Detection Algorithm based on YOLO11-Strawberry

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Abstract

Strawberry is a high-value fruit that is highly susceptible to various diseases due to its delicate skin and environmental sensitivity. Early and accurate detection of strawberry diseases is crucial for minimizing economic losses and ensuring food safety. However, traditional disease detection methods mainly rely on manual visual inspection, which is time-consuming, labor-intensive, and prone to subjective errors. In recent years, deep-learning-based object detection techniques have shown great potential in the agricultural field. This study proposes an improved YOLO11s model, YOLO11-Strawberry, for real-time strawberry disease detection. First, we introduce the VoV-Faster Block, which replaces high-computation modules in the original model to improve computational efficiency while maintaining accuracy. Then, we propose the Multi-Scale Feature Channel Fusion Attention (MSFCA) module, which enhances the detection of small disease regions while reducing background interference. The experimental results demonstrate that YOLO11-Strawberry achieves outstanding performance in strawberry disease detection, with a precision of 93.2%, recall of 92.5%, and mAP50 of 95.3%. Additionally, the proposed improvements lead to a 10.6% reduction in model parameters and a 17.4% reduction in computational cost, significantly enhancing the model's lightweight characteristics and making it suitable for real-time deployment on edge devices. The proposed YOLO11-Strawberry model provides an efficient and lightweight solution for intelligent strawberry disease detection, contributing to the advancement of agricultural automation and precision farming.

Keywords: Strawberry; disease detection; YOLO; computer vision.

Introduction

Strawberry is a high-value and nutrient-rich fruit, abundant in vitamin C, dietary fiber, and antioxidants, which help increase immunity, promote digestion, and improve cardiovascular health (Huang *et al.*, 2023). However, due to their thin skin, fragility, and susceptibility to moisture, strawberries are highly perishable during cultivation, transportation, and storage. Rotten strawberries may carry bacteria and fungi, spreading pathogens through air or contact, thereby accelerating the decay process (Patel *et al.*, 2022). Therefore, improving the efficiency of early strawberry disease detection is crucial for reducing economic losses and ensuring food safety.

Currently, strawberry disease detection mainly relies on manual visual inspection, which is inefficient, subjective, labor-intensive, and prone to visual fatigue over extended periods, leading to an increased rate of missed detections. Additionally, traditional plant disease detection methods are costly and have high false detection rates, making them unsuitable for high-precision and efficient strawberry disease detection. To ensure both accuracy and efficiency in strawberry quality screening, a detection method must meet the requirements for precision and real-time performance. At present, the early detection and control of crop diseases in smart agriculture has become a research hotspot. The era of precision agriculture has undergone a revolutionary transformation through the integration of cutting-edge technologies such as machine learning, deep learning, and the Internet of Things (Gupta and Tripathi, 2024).

With the rapid development of computer vision technology, vision-based detection has been applied to agricultural crop defect detection. Traditional visual detection methods, such as principal component analysis for feature wavelength extraction, have been used to classify four types of epidermal defects in loquats based on principal component (PC) images (Han *et al.*, 2023). Similarly, brightness and texture features extracted from RGB images of ginger have been utilized with Support Vector Machines (SVM), Back Propagation Neural Networks, and Random Forests (RF) for the identification of sulfur-fumigated ginger (Wang *et al.*, 2023b). In another study, deep features of apple images were extracted using Convolutional Neural Networks (CNN) and classified using SVM, RF, *k*-Nearest Neighbors (KNN), Logistic Regression, and Extreme Gradient Boosting (Buyukarikan and Ulker, 2022).

However, traditional vision-based detection requires manual adjustment of model parameters for specific

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detection environments, making it less robust to environmental changes. In contrast, deep-learning-based vision technology enables end-to-end training with greater robustness. For instance, researchers have proposed two CNN models for fruit classification: the first is a lightweight CNN model with six convolutional layers, and the second is a fine-tuned VGG16 model. Both models achieved excellent performance on standard, high-quality test datasets, with classification accuracies of 99.49% and 99.75%, respectively. Additionally, when tested on randomly selected online fruit images, classification accuracies of 85.43% and 96.75% were achieved, demonstrating strong predictive capabilities (Hossain *et al.*, 2018).

Strawberry disease detection is an essential application area for such advancements. For example, a Faster-RCNN-based multitask learning model has been proposed to detect strawberry petioles and leaves and infer whether the entire plant is infected with Verticillium wilt. This model achieved an average detection accuracy of 77.54% on a dataset containing four disease categories (Nie *et al.*, 2019). Similarly, an improved SSD network incorporating the Inception module and replacing the Inception Convolution Kernels with the MEAN module achieved 83.12% accuracy in detecting apple leaf diseases (Liu *et al.*, 2022). Although these methods have shown detection capability, their recognition accuracy remains suboptimal.

In deep-learning-based object detection, YOLO-based or single-stage approaches are among the most successful algorithms, and they are also widely used in plant leaf disease detection (Pacal *et al.*, 2024). Models based on the You Only Look Once (YOLO) (Redmon, 2016) architecture exhibit strong performance. For example, improvements to the YOLOv3 network—such as multiscale feature fusion based on image pyramids, anchor box dimension clustering, and multiscale training—achieved a detection accuracy of 92.39% on a self-constructed multiclass tomato pest and disease dataset (Liu and Wang, 2020). Another study improved the YOLOv4 model to identify powdery mildew on strawberry leaves by replacing the original backbone and neck networks with deep convolutional and hybrid attention mechanisms, effectively reducing the model size (Li *et al.*, 2022b). The DFN-PSAN model was also employed to detect various plant diseases across the PlantVillage, BARI-Sunflower, and FGVC8 datasets, achieving an accuracy of 95.27%, which demonstrates its effectiveness across diverse datasets (Dai *et al.*, 2024).

Among the YOLO series (Redmon and Farhadi, 2017; Bochkovskiy *et al.*, 2020; Li *et al.*, 2022a; Wang *et al.*, 2023a), YOLO11 has gained wide recognition across various fields. YOLO11, introduced by the Ultralytics team in 2024, is the latest object detection model optimized in terms of architecture, efficiency, and task adaptability. As a highly efficient single-stage object detection algorithm, YOLO11 offers high inference speed and accuracy. Designed as an end-to-end model, YOLO11 performs object classification and localization in a single forward pass, transforming object detection into a regression problem that directly predicts the bounding box coordinates and class labels in an image. However, its practical performance requires further innovation and optimization based on specific application scenarios. However, due to the high cost of graphics cards, their application in outdoor domains such as agriculture is often constrained by requirements for high power consumption, substantial heat dissipation, and elevated costs, making them unsuitable for practical deployment. As a result, edge devices are generally more cost-effective. Nevertheless, edge devices such as the Jetson Xavier NX can only achieve 21 frames per second (FPS) when running YOLOv5s, which fails to meet real-time performance requirements. Therefore, it is essential to reduce the computational complexity of the model to ensure real-time inference capability.

Based on the analysis of the actual environment for strawberry disease detection, the following issues need improvement. First, due to varying shooting distances, the strawberry target appears either too large or too small in the image, resulting in multiscale targets, which makes target recognition difficult. Second, the disease range, shape, and color of strawberries vary, and different diseases may present similar color textures, making it challenging to identify the categories of strawberry diseases. Third, detection models often need to be deployed on edge devices, which require lightweight and real-time performance. Therefore, improving the YOLO11 model to detect strawberry diseases is a reasonable approach.

In this study, an improved model based on YOLO11, called YOLO11-Strawberry, is proposed and optimized for strawberry disease detection tasks to increase detection accuracy and meet the real-time requirements of edge devices. The main contributions are as follows:

1. The VoV-Faster Block module is designed to replace high computational modules in the original model, achieving lightweight optimization while maintaining high detection accuracy and reducing computational costs.
2. An MSFCA module is proposed to enhance the detection capability of small-target disease areas while also reducing the interference of complex backgrounds on the model's recognition performance.
3. The experimental results demonstrate that the improved model, YOLO11-Strawberry, outperforms the existing mainstream model YOLO11s in strawberry disease detection ability and is capable of meeting the detection needs for strawberry diseases.

Materials and Methods

Date set

The image data used in this study were collected through two methods: web crawling and manual photography in real-world environments, with the latter performed using a smartphone. To ensure the diversity of the dataset, no fixed lighting conditions were imposed during the image capture process, thereby reducing the model's dependency on lighting variations. All the images were carefully screened and filtered to remove those that were blurry or unclear, aiming to reflect strawberry disease manifestations that may occur under practical conditions as accurately as possible. Each image was manually annotated to ensure dataset quality and usability. To standardize image dimensions, all the images were resized to 608×608 pixels.

A total of five categories were defined: Phytophthora Fragariae (red stele root disease), Anthracnose Fruit Rot, Gray Mold, Powdery Mildew, and healthy fruit, comprising a total of 900 images. Disease identification was based on

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visual inspection by humans. The final annotation statistics included 159 instances of Phytophthora Fragariae, 236 instances of Anthracnose Fruit Rot, 447 instances of Gray Mold, 183 instances of Powdery Mildew, and 1277 instances of healthy fruit.

The dataset was randomly divided into training (700 images), validation (100 images), and test (100 images) sets. In order to better understand the dataset, the sizes of diseased strawberry targets were analyzed. Figure 1 illustrates the relative width and height ratios of the detection targets to the original images, where darker colors indicate higher density. It can be observed that most targets are small relative to the original image, although some large targets are also present. This suggests that it is necessary to accommodate objects of varying scales and achieve multiscale recognition.

YOLO11

YOLO11 is a real-time object detection model released by the Ultralytics team in 2024. It continues the single-stage detection architecture of the YOLO series and includes five versions: n, s, m, l, and x. All versions share the same network structure, differing only in network width and depth. In general object detection tasks, YOLO11n has the fastest recognition speed but the lowest detection accuracy, YOLO11x has the highest detection accuracy but the slowest recognition speed, and YOLO11s strikes a balance between recognition speed and accuracy.

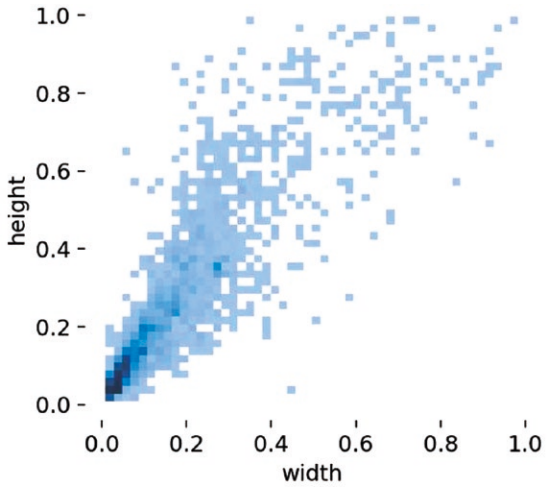


Figure 1. Width and height of the bounding box.

Since strawberry disease detection is often performed on embedded devices with limited computational power but high accuracy requirements, this paper focuses on improving the model based on YOLO11s. The model structure is shown in Figure 2 and consists of three parts: Backbone, Neck, and Prediction. The Backbone part includes the CBS module, C3KA module, C3KB module, Spatial Pyramid Pooling Fast (SPPF) module, and C2PSA module. The CBS module is a conventional convolution module consisting of a convolution layer, batch normalization layer, and Swish activation function. The C3KA module passes part of the input features through a residual connection for direct transfer, while the other part undergoes feature extraction through multiple Bottleneck modules. The C3KB module is similar to the C3KA module but replaces the Bottleneck module with the C3K module. The SPPF module is used to enhance the receptive field by performing pooling operations on the feature map at different grid sizes, enabling multiscale pooling and fusion of features. The C2PSA module, composed of a convolution module and the Pyramid Squeeze Attention (PSA) structure, is used to improve the feature extraction capability of the model. The Neck part combines features from different levels to aggregate high- and low-level features of the image, enhancing the effectiveness of object detection. Finally, the model performs object prediction and output through the Prediction part.

Model lightweight optimization

Although the YOLO11s network has fewer parameters and higher recognition accuracy compared with the previous object detection models, the number of model parameters in deep learning only determines the model's memory usage and

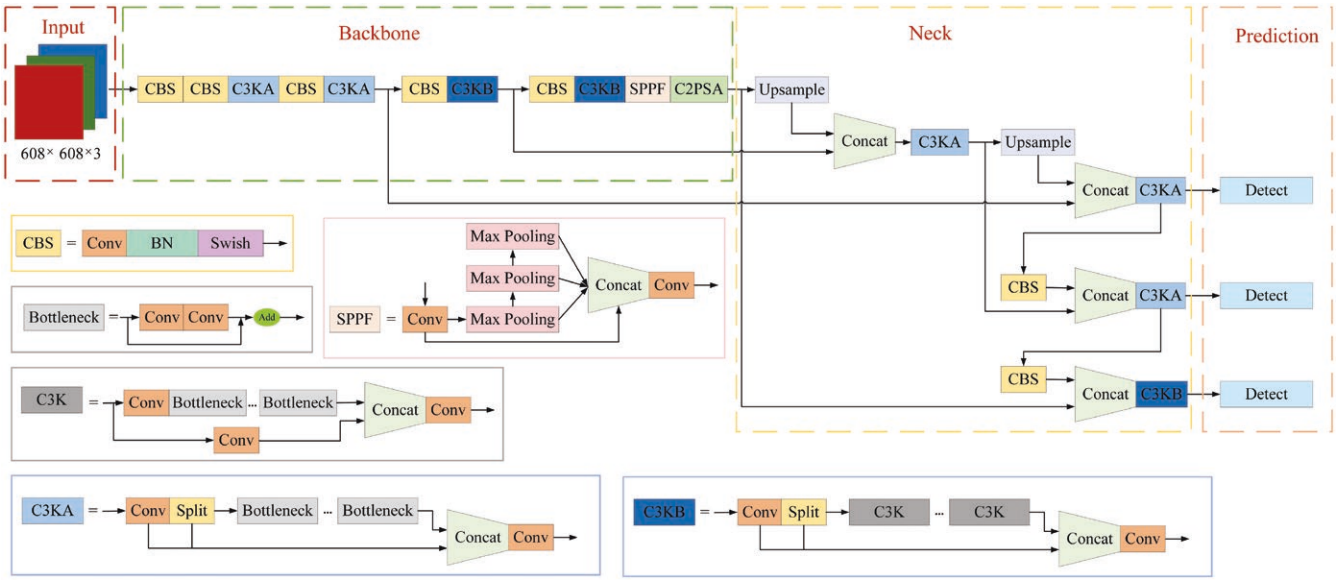


Figure 2. YOLO11 model structure.

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complexity but does not directly represent the running speed. Therefore, to meet the recognition speed requirements of edge devices, it is necessary to improve the model's computational load while maintaining a small number of parameters.

FasterNet is a novel neural network architecture designed to increase computational speed and efficiency, demonstrating particularly strong performance in computer vision tasks. To this end, FasterNet introduces the Partial Convolution (PConv) (Chen *et al.*, 2023) operator, whose model structure is illustrated in Figure 3A. PConv applies convolution operations to only a subset of the input feature map channels, leaving the remaining channels unchanged. This approach reduces computational redundancy and memory access, thereby improving overall efficiency.

Although traditional methods such as depthwise convolution (DWConv) (Chollet, 2017) and group convolution (GConv) (Xie *et al.*, 2017) are effective at reducing floating-point operations, they often result in increased memory access, which leads to higher actual latency and does not necessarily accelerate inference speed. Similarly, the lightweight GhostNet model requires the generation of partial features through standard convolution before performing cost-effective operations, which also involves extensive memory access. This reduces inference speed, and to some extent, the simple linear operations in the Ghost module may result in information loss.

As shown in Figure 3B, the FasterNet Block consists of one PConv layer and two pointwise convolution (PWConv) layers, with a residual connection added. This design reduces computational complexity while preserving feature diversity and transmission capability.

VoV-GSCSP (Li *et al.*, 2024) is a module designed using a one-time aggregation method aimed at reducing computational complexity and network structure complexity while maintaining sufficient accuracy. This module performs effective information fusion between feature maps at different stages, enhancing the model's feature representation ability. In the field of object detection, VoV-GSCSP is commonly used to replace traditional bottleneck modules to accelerate the inference process. In this paper, VoV-Faster is designed by combining the FasterNet Block and VoV-GSCSP structure to reduce the model's computational load and accelerate model inference speed while maintaining model recognition accuracy. The model structure is shown in Figure 4. VoV-Faster is used to replace the C3KA and C3KB modules in the Neck part of the YOLO11s model.

Model recognition accuracy optimization

From the analysis of the dataset, it is evident that strawberry targets exist at multiple scales and shapes, and the model needs to perform multiscale recognition. In practical applications, the detection environment is complex

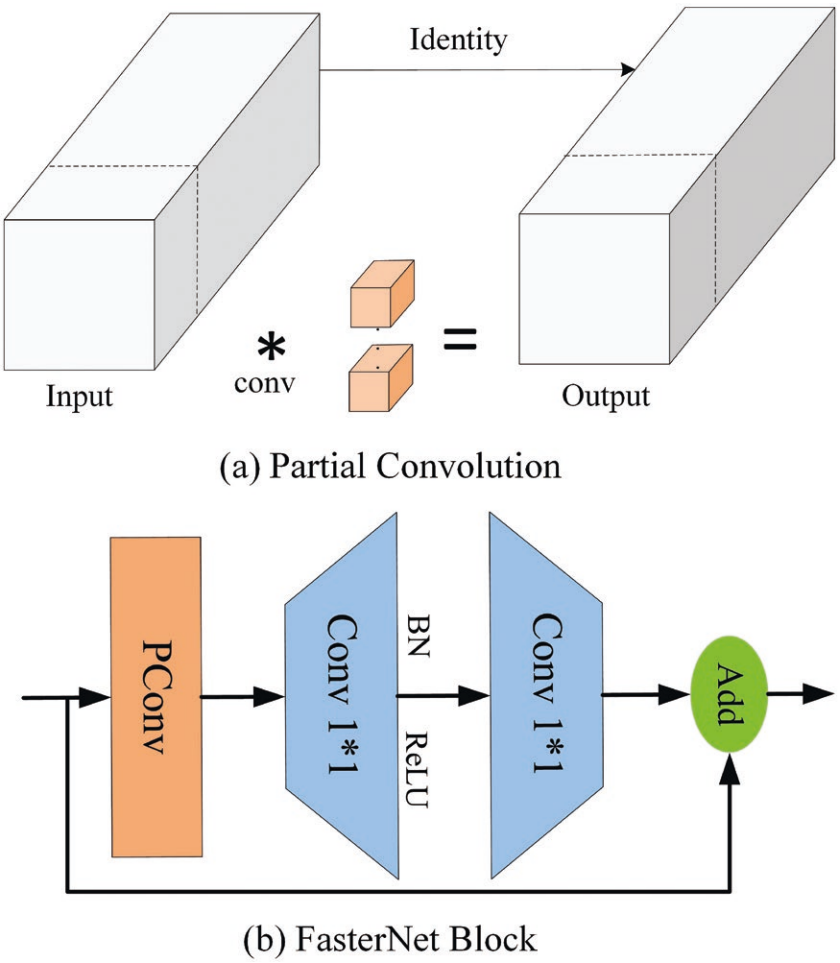


Figure 3. FasterNet Block model structure.

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and variable, with a large number of small-target diseases in strawberries. In deep neural networks, the feature information of small targets tends to be overwhelmed layer by layer, leading to information loss. To improve the model's ability to extract strawberry disease features, this paper designs an MSFCA module to extract features at different scales, filter out irrelevant background features at multiple levels, and effectively fuse the features along the channel dimension. This allows for the efficient extraction of target features and improves the model's recognition accuracy.

The model structure is shown in Figure 5. First, the input features are split along the channel dimension, dividing them into three separate features. Each of the three features undergoes corresponding combinations and convolutions, and then they are recombined along the channel dimension. The overall features are then subjected to convolution for deep feature extraction. Additionally, a Mean Pooling module is designed to apply attention weighting along the channel dimension. This ultimately achieves multiscale and multidimensional feature fusion of the input features.

YOLO11-Strawberry

Based on the aforementioned improvements, the final structure of the improved YOLO11-Strawberry model is illustrated in Figure 6. In this architecture, the C3KA and C3KB modules in the original YOLO11s model are replaced with VoV-Faster modules, and MSFCA modules are integrated into the backbone network for feature extraction. The purpose of incorporating the MSFCA module is to mitigate the progressive loss of small-object feature information during network propagation, extract features at multiple scales, filter out irrelevant background features across multiple levels, and effectively fuse features along the channel dimension. This enables efficient extraction of target features and enhances model detection accuracy.

To this end, two MSFCA modules are inserted into the intermediate layers of the backbone to strengthen the representation of small- and medium-sized objects. Additionally, another MSFCA module is added in the early stage of the neck network to further amplify features. Following this, operations such as upsampling are performed, and detection results for small objects are produced.

Experimental environment

The configuration used in the experiments is as follows: CPU, Intel® Xeon® Silver 4210R (2.40 GHz); Memory, 128 GB; GPU, NVIDIA RTX4090 24G; Operating System, Ubuntu 20.04; Deep Learning Framework, PyTorch 2.1.0; and CUDA Version, 12.1. During model training, the input image size is set to 608×608, the batch size is 40, the epochs are 250, the initial learning rate is 0.01, the weight decay rate is 0.0005, and the momentum is set to 0.937.

Model evaluation

To validate the effectiveness of the improved model, the model's real-time performance is measured using the number of floating-point operations per second (GFLOPS), the number of model parameters (Np), and the FPS. The model's recognition accuracy is measured using Precision (P), Recall (R), Average Precision (AP), and the mean Average Precision (mAP) for each category (Zhu *et al.*, 2019) with the following calculation formulas:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{1}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2}$$

$$\mathrm{AP} = \int_0^1 \mathrm{P(r)}\, \mathrm{d}r \tag{3}$$

$$\mathrm{mAP} = \frac{\sum \mathrm{AP}}{N} \tag{4}$$

In the above formulas, TP refers to true positives (positive samples correctly predicted as positive), TN refers to true negatives (negative samples correctly predicted as negative), FP refers to false positives (negative samples incorrectly predicted as positive), and FN refers to false negatives (positive samples incorrectly predicted as negative). AP is defined as the area under the Precision–Recall (P–R) curve generated at different confidence thresholds, where P(r) represents the precision at a given recall value r. N denotes the number of classes. Precision measures the proportion of true disease instances among all instances predicted as diseased by the model. This reflects the model's ability to control false positives, that is, to avoid misdiagnosing healthy strawberries as diseased, thereby preventing unnecessary intervention on healthy fruit. Recall measures the proportion of correctly identified diseased instances among all actual diseased instances. This reflects the model's ability to minimize false negatives, which is crucial in evaluating whether the model can detect the majority of disease cases.

In the context of strawberry disease monitoring, missing severe lesions may lead to disease spread, making high recall particularly critical for early intervention. The AP provides a comprehensive evaluation of the balance between precision and recall. Instead of measuring detection performance at a single threshold, it assesses the model's overall detection capability across varying confidence levels, thereby offering a more holistic view of its stability and robustness. mAP reflects the model's overall classification and detection performance across all disease categories. A high mAP indicates that the model performs well not only for a specific disease type but also for maintaining consistent detection performance across

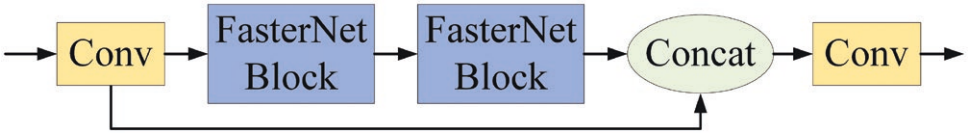


Figure 4. VoV-Faster model structure.

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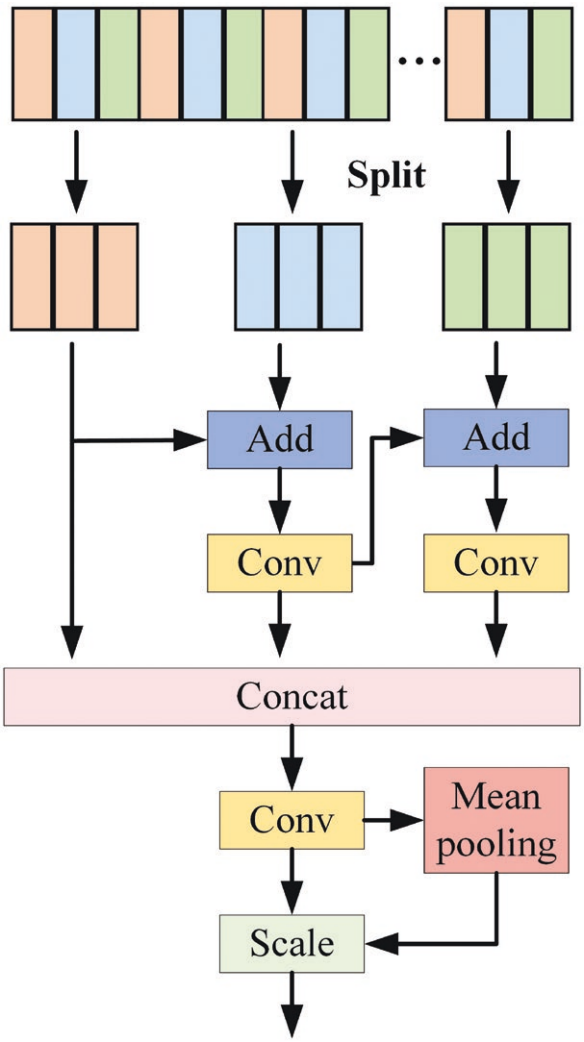


Figure 5. MSFCA model structure.

multiple disease classes, thus demonstrating strong practical application potential.

Results and Discussion

Ablation experiment

To validate the effectiveness of the proposed modules, ablation experiments were conducted based on the YOLO11s baseline algorithm, and the contributions of the VoV-Faster Block and the MSFCA module were independently evaluated. All ablation experiments were performed using the same test set and identical hyperparameters. The training loss, precision, and recall curves of the YOLO11-Strawberry model are presented in Figure 7. It shows that the model converges successfully upon completion of training.

The experimental results are shown in Table 1. Compared with the baseline algorithm, the model using the VoV-Faster Block module reduced the number of parameters by 19% and the computational cost (FLOPS) by 24.4%, which demonstrates that the computational load of the model can be effectively reduced without a decline in overall detection accuracy. This finding validates that the proposed module contributes to making the model more lightweight.

In comparison to the baseline algorithm, the model using the MSFCA module achieved a 2.3% improvement in precision, indicating a reduction in false positive rates and demonstrating the effectiveness of the attention module in filtering out interference information. Recall increased by 3.1%, highlighting the model's enhanced ability to detect target features that are prone to being overlooked. Additionally, the mAP50 (the mAP at an Intersection over Union (IoU) threshold of 0.5) improved by 2.2%, suggesting that the model is more robust. Therefore, the MSFCA module effectively enhances the model's recognition performance.

To further visualize the effectiveness of the MSFCA module, heatmap experiments were conducted to display the regions the model focuses on, with darker colors indicating greater attention. As shown in Figure 8, the YOLO11s baseline model fails to focus well on the disease areas of strawberries and instead pays attention to certain parts of the background, which may cause the model to be distracted by the background environment and decrease its detection accuracy. In contrast, the YOLO11s model with the MSFCA module demonstrates a strong focus on strawberry disease regions, effectively reducing background interference and improving model performance. Thus, the heatmap results also confirm the effectiveness of the MSFCA module.

Finally, we combined the above improvements to create the YOLO11-Strawberry model. Compared with the baseline YOLO11s, this model shows a 1.9% improvement in precision, a 3.8% increase in recall, and a 2.2% improvement in mAP50. The number of parameters decreased by 10.6%, and FLOPS reduced by 17.4%. This finding demonstrates that the model effectively reduces the parameter size, minimizes the impact of complex backgrounds on recognition, and increases accuracy. The improvements led to better target feature representation and increased recall, confirming the effectiveness of the proposed model enhancements.

Comparison experiment

To demonstrate the advantages of the YOLO11-Strawberry improved model, comparative experiments were conducted against the classic networks Faster R-CNN (Girshick, 2015) and SSD (Liu *et al.*, 2016), as well as YOLOv8s and YOLO11m, with hyperparameters held constant. The comparison results are shown in Table 2. As can be seen from the table, compared with the Faster R-CNN and SSD models, the YOLO series models offer significant advantages in both accuracy and detection speed, confirming that selecting the YOLO architecture as the base for the improved model is a sound decision.

The baseline model YOLO11s outperforms both YOLOv8s and YOLOv10s in terms of recognition accuracy and speed, which justifies the selection of the YOLO11 series as the baseline model. Furthermore, compared with the YOLO11n model, YOLO11s exhibits better recognition performance. However, YOLO11m does not meet the real-time detection requirements for edge devices in terms of detection speed, leading to the final choice of YOLO11s as the improved base model.

The experimental results show that, compared with the baseline YOLO11s model, the YOLO11-Strawberry model improves detection speed, reduces model parameters, and enhances detection accuracy, yielding superior performance.

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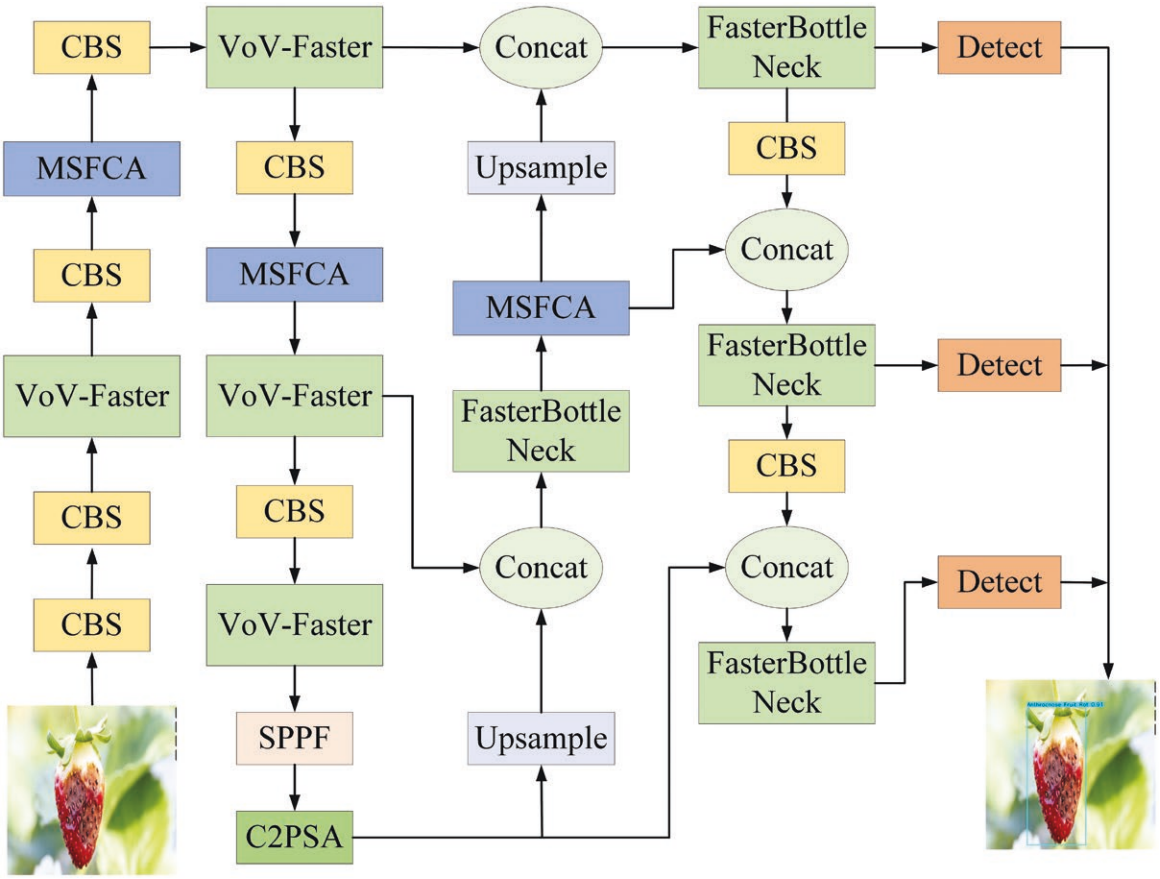


Figure 6. YOLO11-Strawberry model structure diagram.

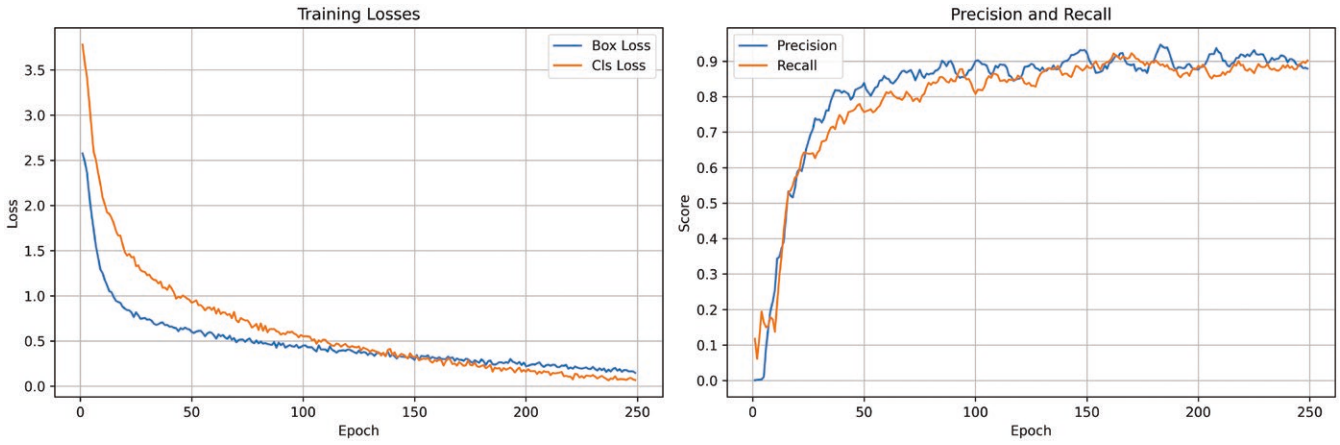


Figure 7. Model training loss, precision, and recall.

Table 1. Result of ablation experiment with YOLO11s adding different modules

Model	Precision/% ↑	Recall/% ↑	mAP50/% ↑	Parameters/M ↓	FLOPS/G ↓
YOLO11s	91.3	88.7	93.1	9.4	21.3
YOLO11s+ VoV-GSCSP	91.2 (−0.1)	89.9 (+1.2)	93.4 (+0.3)	7.6 (−1.8)	16.1 (−5.2)
YOLO11s+MSFCA	93.6 (+2.3)	91.8 (+3.1)	95.0 (+1.9)	10.2 (+0.8)	22.8 (+1.5)
YOLO11-Strawbery	93.2 (+1.9)	92.5 (+3.8)	95.3 (+2.2)	8.4 (−1.0)	17.6 (−3.7)

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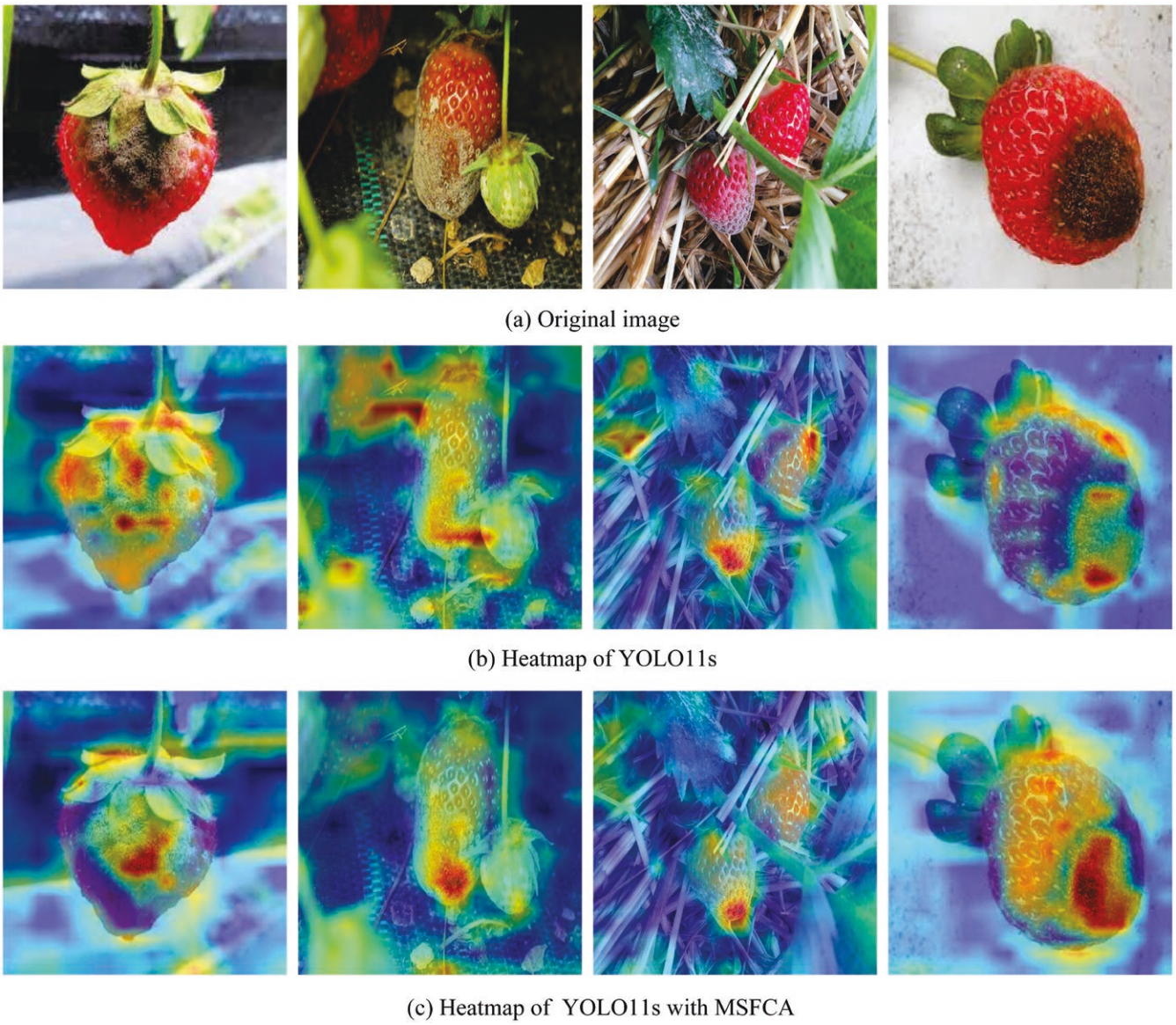


Figure 8. Comparison of heat map.

Table 2. Comparison of experimental results of different target detection algorithms

Model	Precision/% ↑	Recall/% ↑	mAP50/% ↑	Parameters/M ↓	FLOPS/G ↓
YOLOv8s	90.2	89.9	92.6	11.2	28.4
YOLOv10s	90.6	88.6	92.7	9.5	21.7
YOLO11n	85.4	83.8	89.1	2.6	6.3
YOLO11s	91.3	88.7	93.1	9.4	21.3
YOLO11m	92.0	90.4	94.7	20.1	67.9
Faster R-CNN (VGG)	86.9	85.2	87.5	77.3	171.3
SSD (VGG)	85.3	83.6	86.0	43.7	105.2
YOLO11-Strawbery	93.2	92.5	95.3	8.4	17.6

The model achieves a precision of 93.2%, recall of 92.5%, and mAP50 of 95.3%. The model's parameter count is reduced to 8.4M, and the computational complexity is decreased to 17.6G. Additionally, the GPU running speed in this experiment is increased by 23 FPS. For edge devices like the Jetson Xavier NX, YOLO11s runs at 21 FPS, while the YOLO11-Strawberry model reaches 27 FPS, thus meeting the real-time detection requirements for edge devices.

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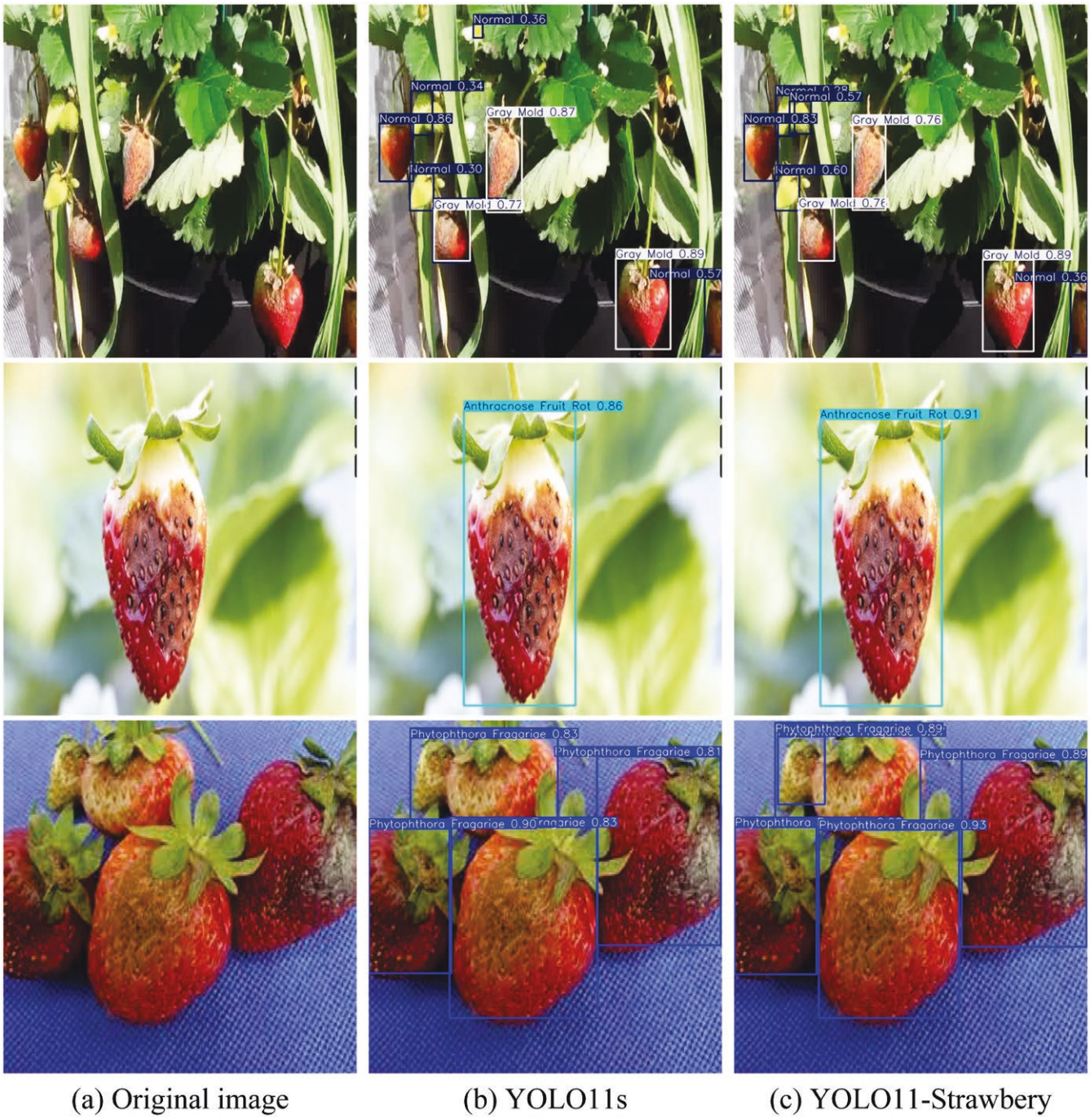


Figure 9. Detection results of YOLO11s and YOLO11-Strawberry.

To further validate the effectiveness of the model, Figure 9 visualizes the detection results of YOLO11s and YOLO11-Strawberry. From the figure, it can be seen that for the first image, YOLOv5s produced a misdetection of the background, whereas YOLO11-Strawberry did not, indicating the improved model's effectiveness in filtering background information. For the second image, which is a large target, both models correctly detected it, but the improved model had a higher confidence score, demonstrating a stronger focus on the target. For the third image, YOLO11 missed a detection, while the improved model showed better attention to the target. Therefore, the experimental results indicate that YOLO11-Strawberry has better detection performance.

Conclusions

This paper proposes a strawberry disease recognition model, YOLO11-Strawberry, based on YOLO11. First, the VoV-Faster Block module is introduced to replace the high computational modules in the original model, achieving model lightweight and effectively reducing the computational load. Additionally, to minimize the interference of background information in model recognition and enhance the model's ability to recognize small-target strawberry diseases, the MSFCA module is added, significantly improving the model's recognition accuracy. The final model achieves a precision of 93.2%, recall of 92.5%, and mAP50 of 95.3%, with parameters of 8.4M

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and a computational load of 17.6G. Compared with the baseline model YOLO11, the model's precision improves by 1.9%, recall by 3.8%, mAP50 by 2.2%, parameters decrease by 10.6%, and FLOPS reduces by 17.4%. Therefore, this study demonstrates that the YOLO11-Strawberry-based strawberry disease recognition algorithm is feasible. The model can effectively filter out interference caused by complex background in real environments, enhance the recognition ability for small-target strawberry diseases, and meet the real-time detection requirements for edge devices.

Author Contributions

Xiaoneng Xie: Conceptualization, methodology, visualization, supervision, project administration, and writing original draft; Ranran Zhang: Conceptualization, methodology, visualization, and review and editing; Jian Guo, Luyuan Lu, Haiqiang Pan, Xu Luo, and Shichen Meng: Conceptualization and methodology. All authors have read and agreed to the published version of the manuscript.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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